CiteBud — Adaptive Study Assistant

ER Diagram & Relationship Cardinalities

DBMS: PostgreSQL 14+ with pgvector | **Tables:** 11

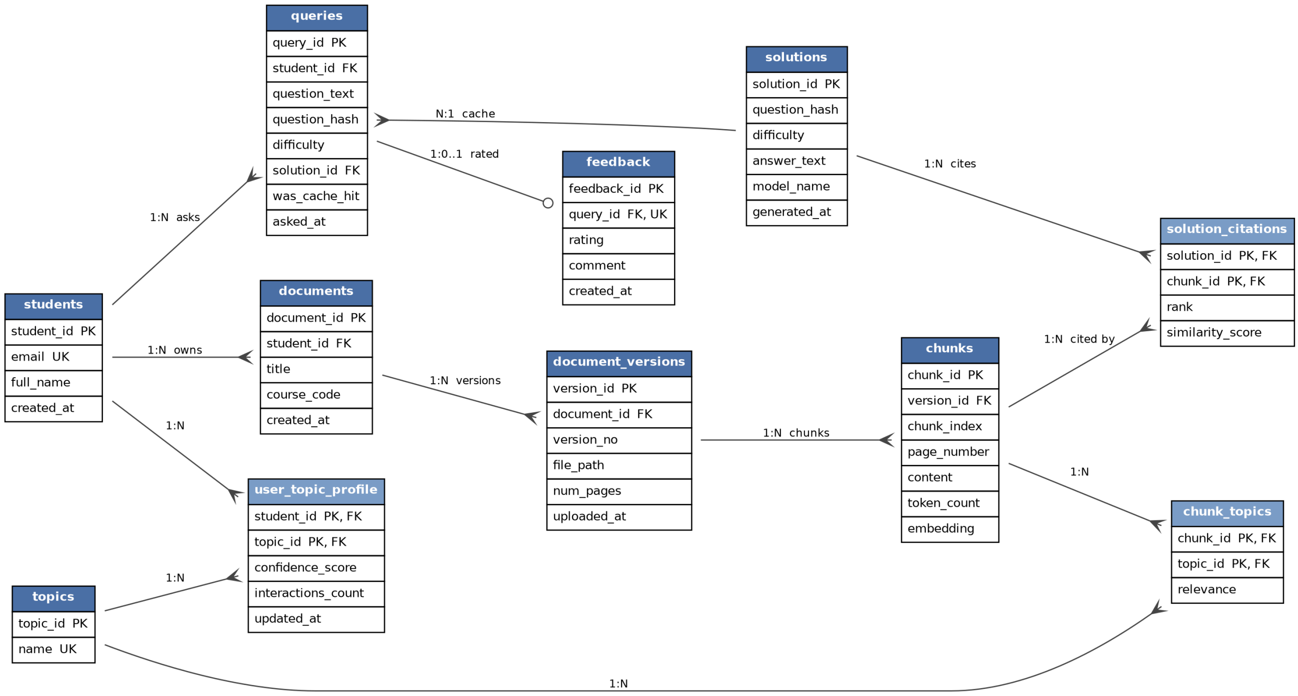

Table Overview

#	Table	Feature it realizes
1	`students`	User identity
2	`documents`	Logical document identity (survives re-upload)
3	`document_versions`	Versioning system — immutable snapshots
4	`chunks`	Retrieval corpus + vector embeddings
5	`topics`	Topic extraction layer (controlled vocabulary)
6	`chunk_topics`	M:N tagging, with per-pair relevance
7	`solutions`	LLM answers, content-addressed for caching
8	`queries`	Question log; points at (possibly shared) solution
9	`solution_citations`	Many-to-many citation tracking with scores
10	`feedback`	Feedback loop — per-query ratings
11	`user_topic_profile`	User-adaptive learning — confidence per topic

Cardinalities Justified

Relationship	Cardinality	Why
students → documents	1 : N	Each upload belongs to exactly one student.
documents → document_versions	1 : N	Re-uploading creates a new version; old versions preserved for reproducibility.
document_versions → chunks	1 : N	Chunking happens per snapshot, so chunks are bound to a version, not a logical document.
chunks ↔ topics	M : N	A chunk can cover several topics; a topic spans many chunks. relevance is a property of the pairing, so it lives in the junction.
students → queries	1 : N	Full history kept.
queries → solutions	N : 1	Caching. Multiple queries with the same normalized hash + difficulty reuse one solutions row; was_cache_hit flags reuse for analytics.
solutions ↔ chunks	M : N	An answer cites several chunks; a chunk can ground many answers over time. rank and similarity_score are properties of the retrieval event, so they live in solution_citations.
queries → feedback	1 : 0..1	At most one rating per query, enforced by UNIQUE(query_id).
students ↔ topics	M : N	Via user_topic_profile — associative entity with its own attributes (confidence, counters).